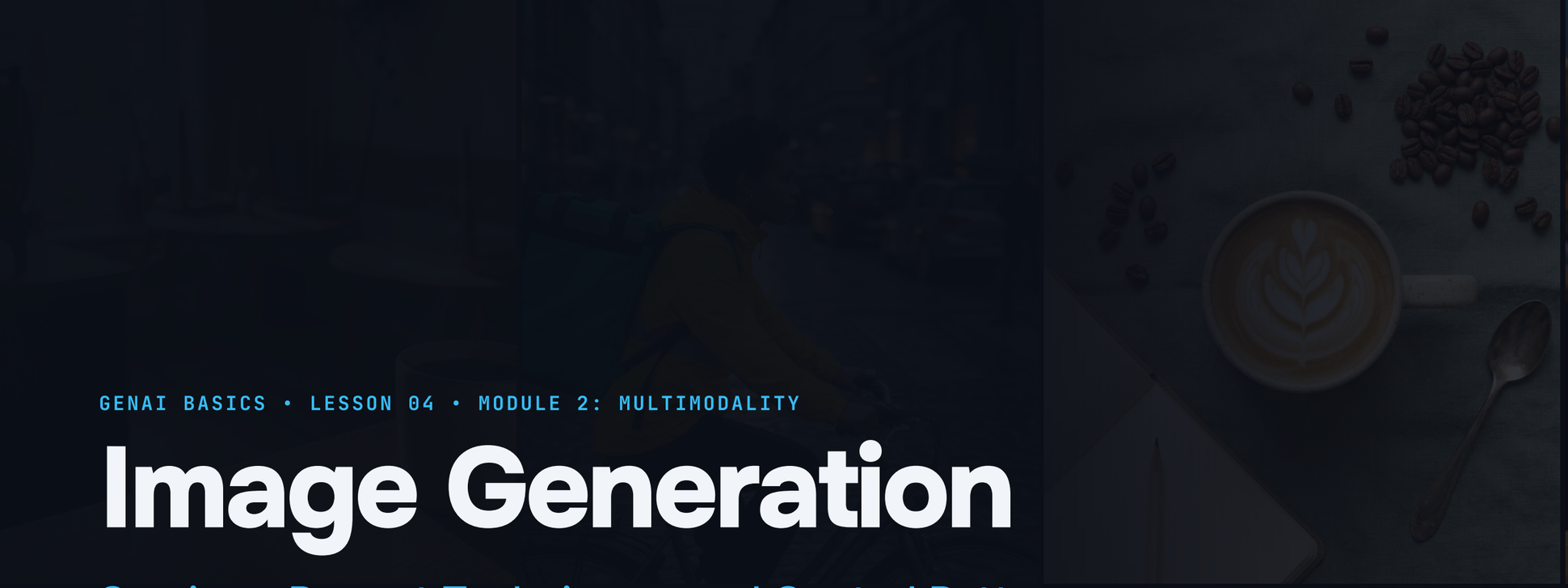
GENAI BASICS • LESSON 04 • MODULE 2: MULTIMODALITY

Image Generation

Services, Prompt Techniques, and Control Patterns

ChatGPT, Nano Banana, Leonardo.ai, Midjourney, Kling, and Sintx. From an interactive quiz and provider architecture to e-commerce packshots, inpainting, and cross-scene character consistency.







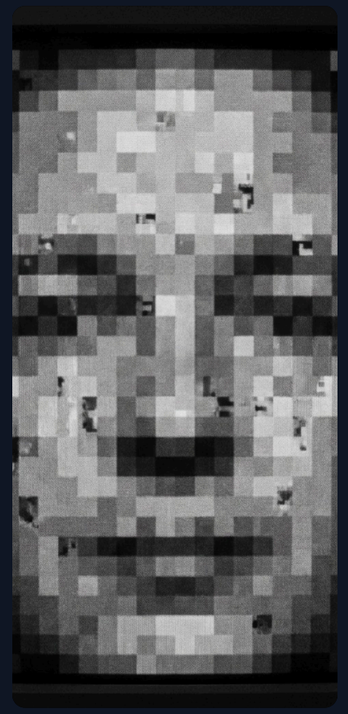


INTERACTIVE HOOK • WARM-UP QUIZ

Guess: Which is a real human, and which is AI?

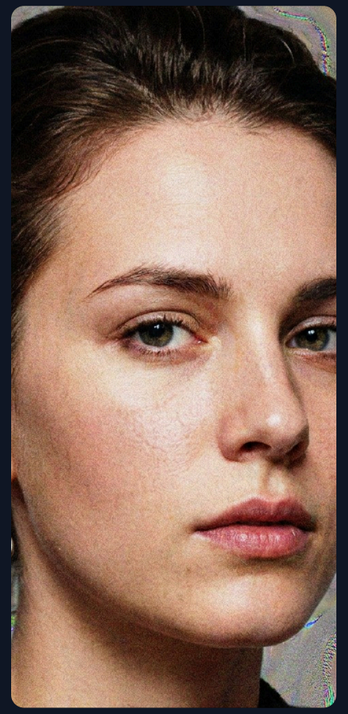
Six portrait images. Post the letters of those you believe are genuine photographs vs. AI generated in chat.

[A]



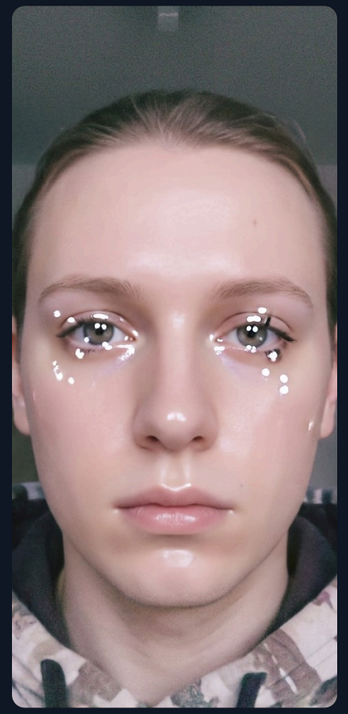
Portrait A

[B]



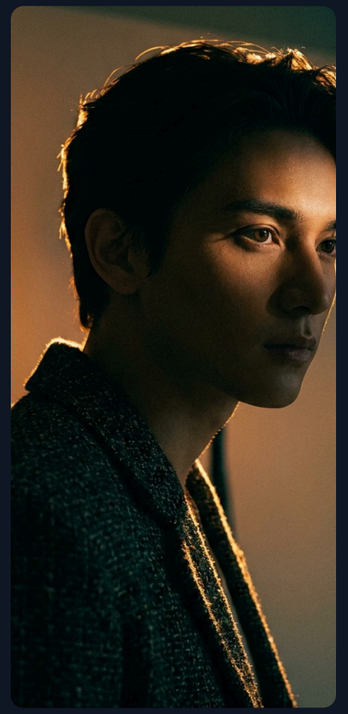
Portrait B

[C]



Portrait C

[D]



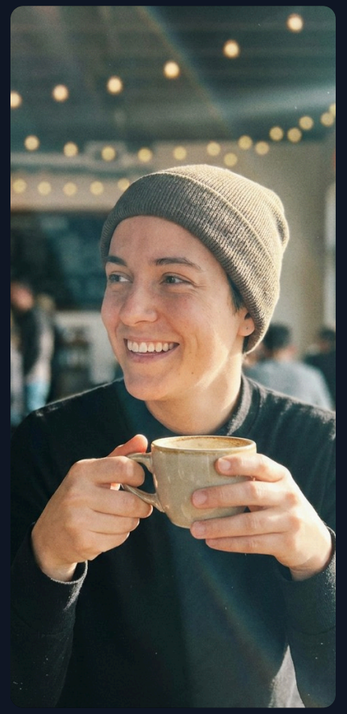
Portrait D

[E]



Portrait E

[F]



Portrait F

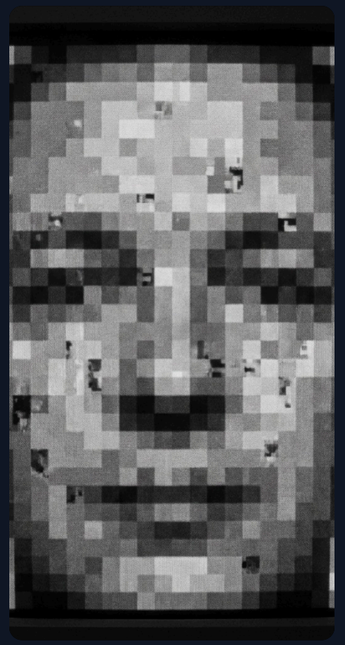
Chat Poll: Enter your guesses. For example: "A and B are AI, E and F are real" or "Only F is real". What visual clues are you looking at?

THE REVEAL • 2014–2026

All 6 are AI generated. 10 years of evolution

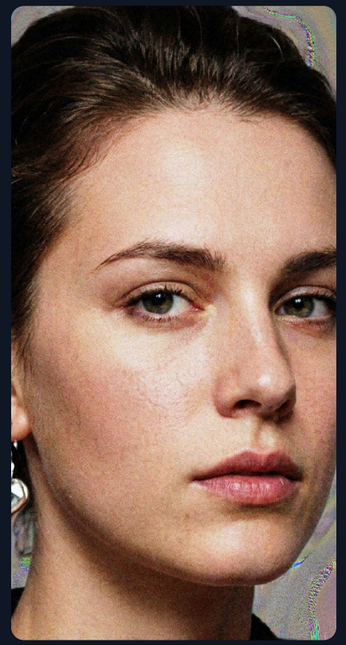
Not a single real photograph. A visual chronicle of how machine learning crossed the uncanny valley in one decade.

2014 GAN



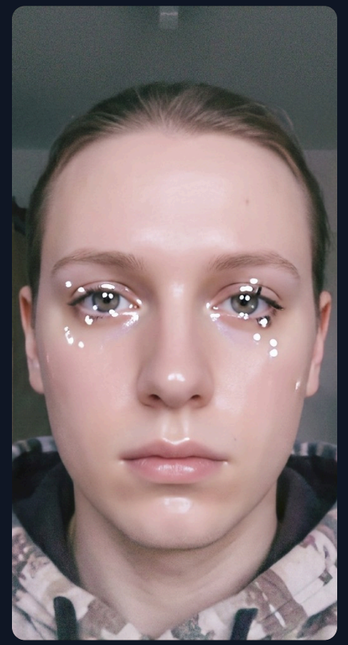
Goodfellow GAN
64×64 px grayscale noise blobs. The birth of deep generative models.

2018 StyleGAN



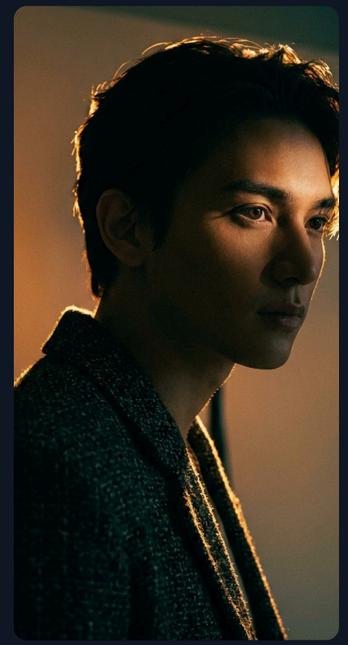
NVIDIA StyleGAN
Early photorealism, but melting earrings, ears, and warped backgrounds.

2022 SD 1.5



Stable Diffusion
Open-source diffusion explosion. Glossy, plastic skin and vacant gaze.

2023 MJ v5



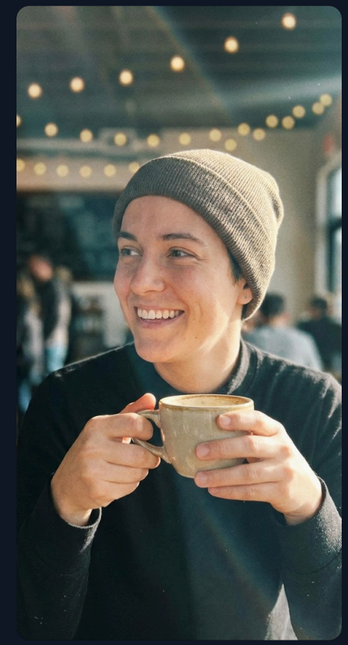
Midjourney v5
Cinematic aesthetic victory. Studio lighting, dramatic chiaroscuro.

2024 FLUX.1



FLUX.1 Dev/Pro
Authentic skin pores, micro-imperfections, natural teeth and anatomy.

2026 Pro VLM



Nano Banana Pro
Organic smartphone optical blur, candid scene, sensor grain. Forensic match.

> **Engineering Takeaway:** Modern models generate beauty effortlessly. The central problem of 2026 is **controllability**: how to steer models to generate exact specifications and maintain subject identity across 10 scenes.

COURSE ROADMAP

Module 1 Completed. Entering Pixels

01 **Foundations & API**
L01–L03

02 **Multimodality**
L04 Images
L05 Video • L06 Audio

03 **Engineer Environment**
L07–L09

04 **Local AI**
L10

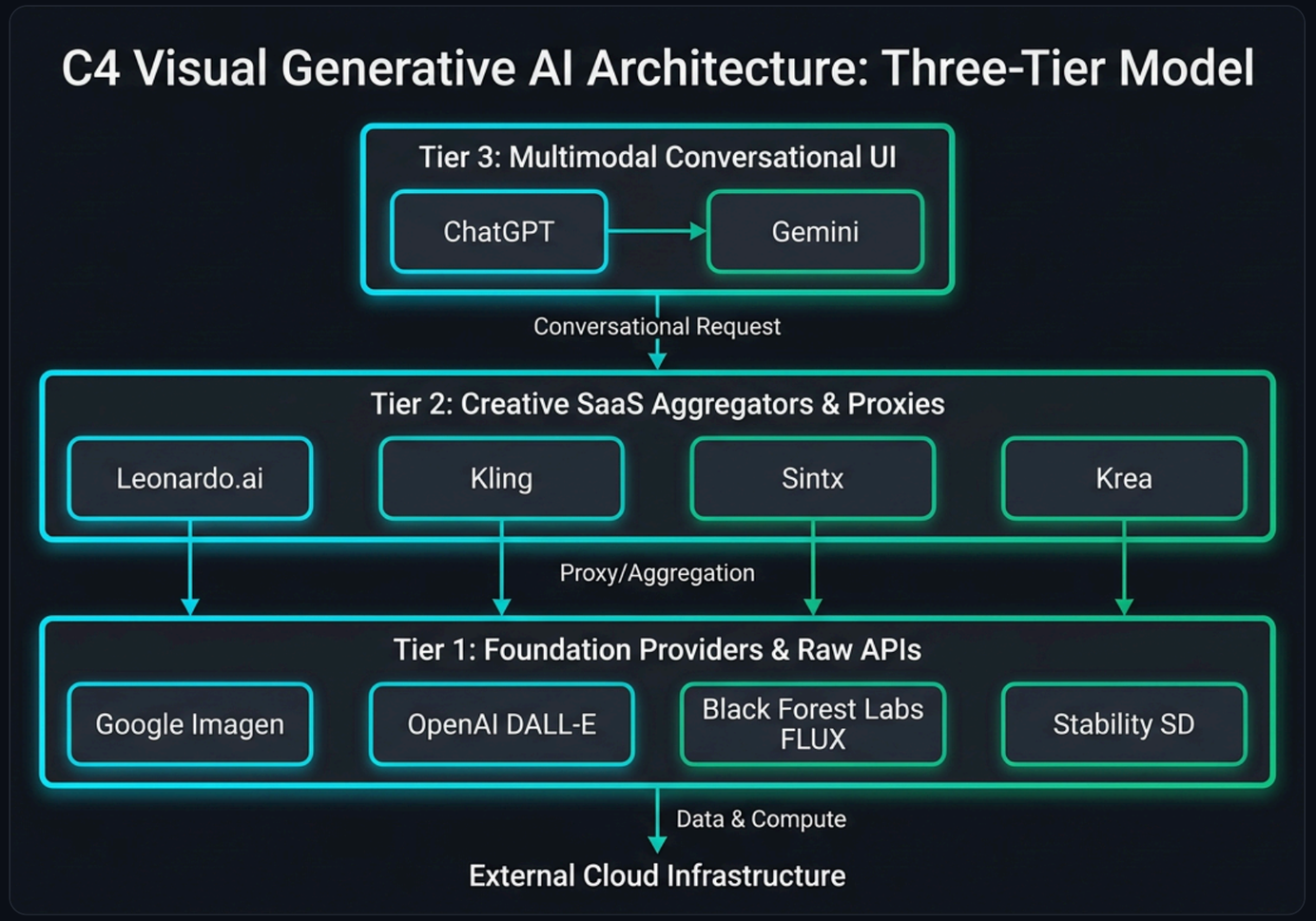
05 **Agents & MCP**
L11–L13

06 **Production**
L14–L16

> **Carried over from Module 1:** Prompt engineering principles from L02 apply directly to vision models. Specificity beats buzzwords, slot separation prevents collision. But instead of text few-shots, we now supply **visual reference images**.

Three Tiers of the Visual AI Ecosystem

How the image generation landscape is structured: from raw foundation weights to creative studios and conversational chats.



Tier 3: Multimodal Conversational UI

ChatGPT, Gemini, Grok Imagine. Unified model reasons over text and vision, generates pixels, and remembers multi-turn dialogue context.

Tier 2: Creative SaaS Aggregators (Proxies)

Leonardo.ai, Kling, Sintx, Krea. Proxy platforms providing an interactive canvas, layers, inpainting brush masks, upscalers, and multi-model routing.

Tier 1: Foundation Model Providers & APIs

Google Imagen, OpenAI DALL-E, FLUX, Stability SD. Compute heavyweights training base weights and exposing raw REST/Python SDK endpoints.

TIER 1 • FOUNDATION PROVIDERS

Who Trains Foundation Models & Exposes APIs

The same infrastructure providers you accessed in L03 via Python SDK.

GOOGLE CLOUD

Imagen 3 / Nano Banana Pro

Models: `gemini-3-pro-image`, `gemini-3.1-flash-image`.
Strengths: Multi-image reference binding, deep semantic comprehension, Python SDK.

OPENAI

DALL-E 3 / GPT Image 2.5

Models: `dall-e-3`, `gpt-4o-image`.
Strengths: Complex in-image typography, strict prompt instruction adherence.

BLACK FOREST LABS

FLUX.1 (Pro / Dev / Schnell)

Available via Replicate, fal.ai, Together AI.
Strengths: Industry benchmark for human skin texture, natural anatomy, open weights.

STABILITY AI

Stable Diffusion 3.5 / SDXL

Open-weights ecosystem staple. Local hosting ready (L10).
Strengths: Thousands of community fine-tunes and custom style LoRAs.

MIDJOURNEY

Midjourney V6.1 / V8.2

Closed proprietary web & Discord suite. No public REST API.
Strengths: Unrivaled commercial and editorial cinematic aesthetic.

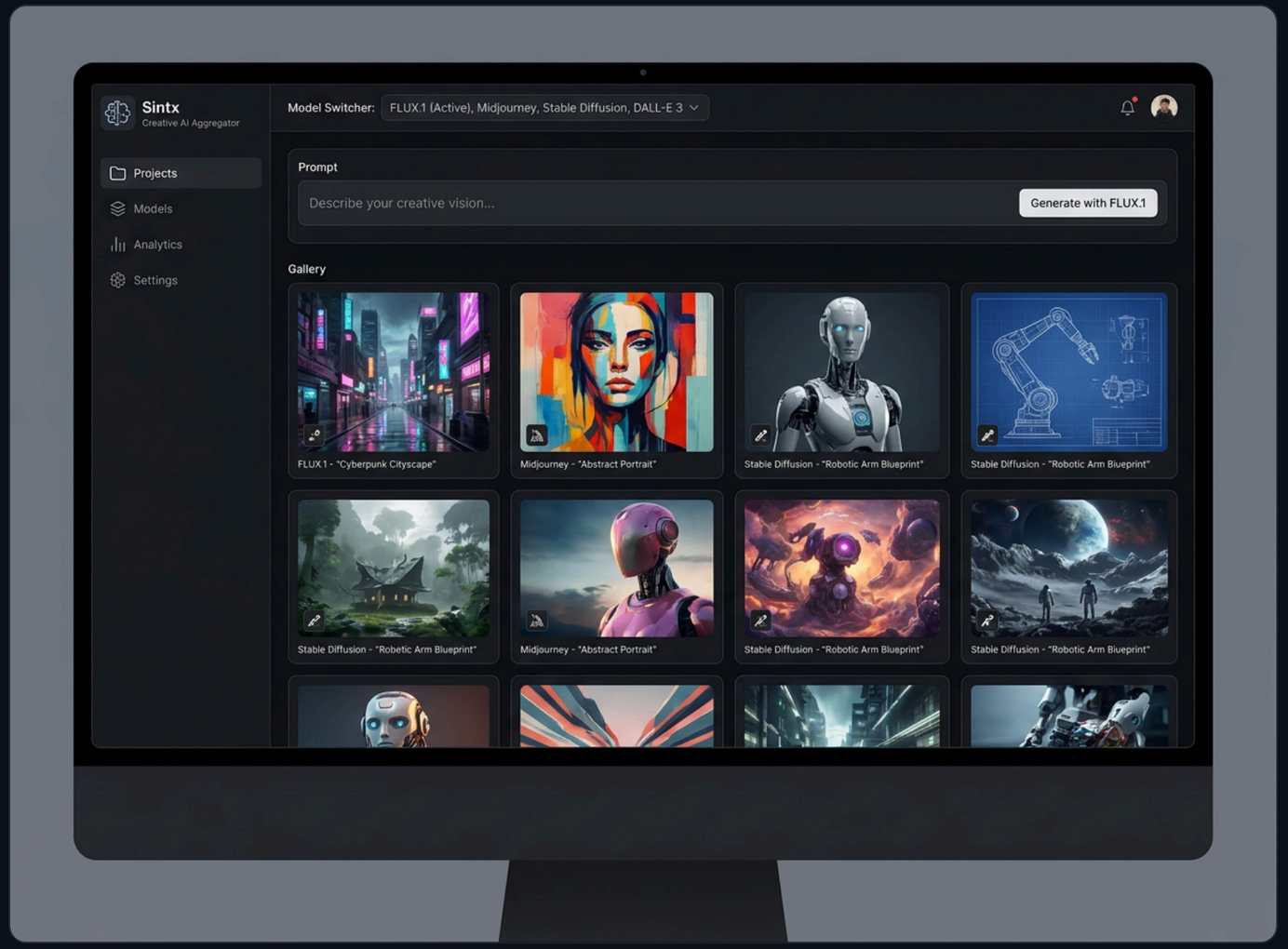
PROGRAMMATIC ACCESS

Direct API Endpoints

`client.models.generate_content` or REST endpoints.
Economics: $0.02 – $0.08 per image. Latency: 3–15 seconds.

Why Use a SaaS Proxy Over Raw API?

Leonardo.ai, Sintx, Kling, Krea: why designers and product teams prefer aggregators.



- **Multi-Model Routing:** One prompt bar allows switching between FLUX, Midjourney, and SDXL without managing 5 different API bills.
- **Interactive Canvas:** Moveable layers, brush-based inpainting masks, and outpainting canvas expansion.
- **Style Presets & LoRAs:** Instant brand styling (corporate 3D, product photoshoot, anime) without writing 100-word prompt tails.
- **Control Parameters:** Fine-tuned sliders for Guidance Scale, Steps, Seed locking, and ControlNet edge/pose detection.

> **Definition:** A creative SaaS is an intelligent proxy wrapping foundation APIs into high-productivity visual workflows.

WORKFLOW COMPARISON • TWO PARADIGMS

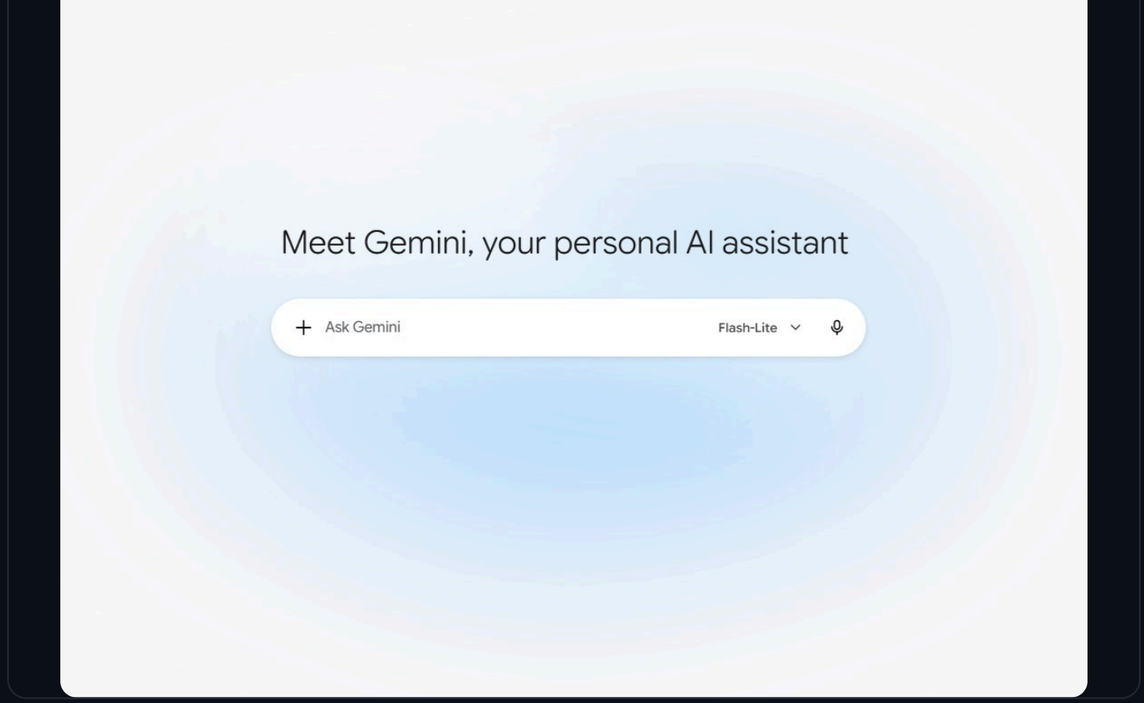
Canvas & Sliders vs. Conversational Semantics

Contrasting the studio graphics workstation against the multimodal conversational LLM.



Leonardo.ai (Creative SaaS)

Paradigm: Graphic Workstation. You tweak diffusion mechanics: Guidance Scale, Negative Prompts, Seed lock, LoRA strength. To edit, you paint a manual brush mask over pixels.



ChatGPT & Nano Banana (Multimodal Chat)

Paradigm: Conversational Director. The model possesses innate vision and spatial reasoning. Edits are natural language instructions: *"Replace only the mug with an iced latte, preserve lighting and table geometry."*

ENGINEERING EVALUATION • SELECTION MATRIX

Which Tool Fits Your Production Need?

EVALUATION METRIC	CREATIVE SAAS (LEONARDO, SINTX)	MULTIMODAL CHAT (CHATGPT, GEMINI)	RAW API (PYTHON SDK / CLOUD)
Entry Barrier	Moderate (layers, sliders, canvas)	Zero (plain natural language)	High (code, auth tokens, pipelines)
Pixel Modification	Maximum (brush mask, canvas)	Semantic (conversational instruction)	API Mask (Base64 / inpainting payload)
In-Image Typography	Fair (depends on selected model)	Superior (ChatGPT, Nano Banana)	Superior (via DALL-E 3 / Imagen 3)
Character Consistency	Character Reference, Seed lock	Multimodal Context (up to 14 images)	Passing reference images in `contents`
Batch Scale (500+ items)	Impractical (manual UI clicks)	Impossible (interactive chat)	Optimal (script, $0.03/card, DB batch)

> **Architecture Rule:** Rapid prototyping and typography ➔ **Chat**. Detailed artistic touch-ups with brushes ➔ **SaaS**. Automated production asset pipeline ➔ **API Script**.

Six Words — Model Makes Every Creative Decision



COMPLETE VAGUE PROMPT

```
a cozy coffee shop in the morning
```

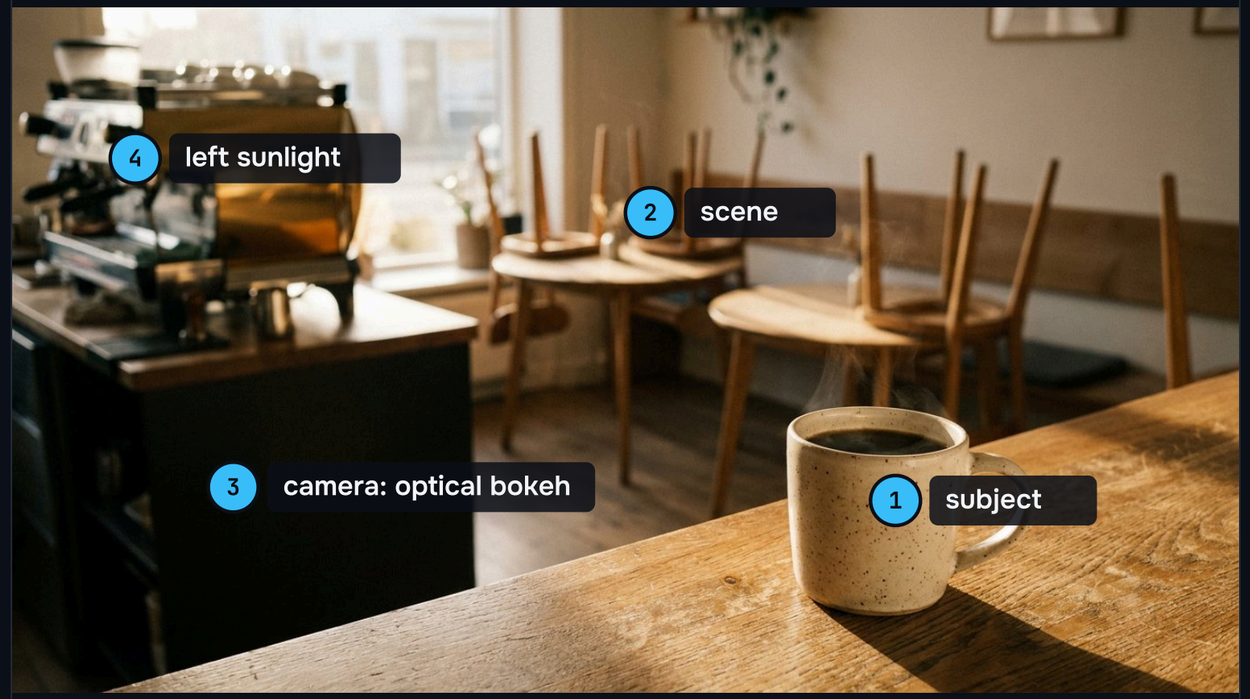
Morning? Outside it's raining. Strangers, a cat, and a fireplace were never requested — the model filled empty latent space with training data averages.

Six Slots — What Controls What and How to Refine

RUNNABLE SLOT PROMPT (COPY & RUN) **100% SCENE CONTROL**

Subject: a speckled cream ceramic mug of black coffee with steam, on worn oak counter in right third.
Setting: empty Scandinavian coffee shop at 7 a.m., chairs up on tables, brass espresso machine on left.
Camera: eye-level, 35mm lens, f/2.0, shallow depth of field.
Lighting: low warm morning sunlight from left window, long soft shadows.
Style: realistic editorial photography. **Constraints:** no people, no text, no logos.

VISUAL INSPECTOR:



SLOT ANATOMY: WHAT EACH ELEMENT CONTROLS IN THE MODEL

1. Subject: Semantic center, texture, rule of thirds (right third).

2. Setting: Time (7 a.m.) & context — eliminates random crowds & baristas.

3. Camera: 35mm (zero distortion), f/2.0 (optical bokeh vs AI blur).

4. Lighting: Low morning rays from left create 3D volume, shadows & grain.

5. Style: Editorial photo blocks plastic 3D gloss & synthetic CGI.

6. Constraints: Guardrails prevent hallucinated letters, logos, and faces.

💡 3 KEY VECTORS TO UPGRADE A WEAK PROMPT

BEST PRACTICE

1. Physical Markers over Vague Adjectives: Avoid "cozy" (model invents cats & fireplaces). Specify concrete geometry: *«7 a.m., chairs up on tables»*.

2. Real Optics over Junk Buzzwords: Remove noise "4k, masterpiece, trending" (ignored by model). Dictate lens physics: *«35mm lens, f/2.0, shallow depth of field»*.

3. Explicit Guardrails: Don't rely on luck. Explicit block *«Constraints: no people, no text, no logos»* eliminates unwanted faces & pseudo-letters.

Critical Details & Typography as Explicit Blocks




```
Must-have:
• small chalkboard sign with text exactly "FLAT WHITE 3.50" and "OPEN 7:00"
• one fresh butter croissant behind the mug
• small glass vase with fresh lavender on left
```

- **Typography:** wrap text in quotes and use the keyword `exactly`.
- **Quantities:** specify numerals ("one croissant", not "croissants").
- **Spatial Anchors:** declare relative positioning ("behind mug", "to left").
- **Negative Bounds:** state `no other text, no typos` explicitly.

Text, croissant, and lavender rendered strictly as specified

One Subject — Six Lens and Light Coordinates



`low angle` heroic perspective



`top-down` flat lay arrangement



`24mm wide` subject + environment



`85mm f/1.8` portrait optical bokeh



`golden hour` warm backlight



`softbox studio` balanced commercial light

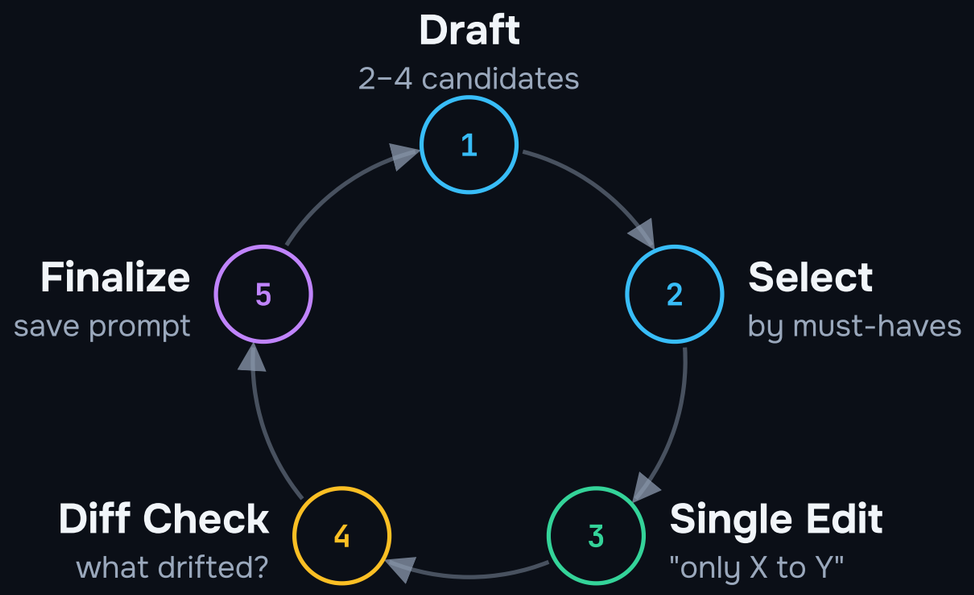
LECTURE EQUATOR

5-Minute Break

05:00

Coming next: surgical inpainting, consistent courier character, and commercial product photoshoot.

Iterate in Loops — Never Reroll the Whole Dice



ENGINEERING REFINEMENT TEMPLATE

```
Edit this image. Modify only object X to Y.
Preserve: angle, lighting, background, and shadows.
Do not change: surrounding room geometry.
```

- **One edit per step** (equivalent to an atomic git commit).
- **Explicitly declare protected regions** to preserve context.
- **After 6–8 edits** — drift accumulates. Start fresh chat using best frame.

One Master Frame — Three Targeted Edits



Original A2: morning coffee shop, ceramic mug



Object (B1): mug ➔ iced latte



Lighting (B2): rainy evening outside window



Style (E2): flat 2D vector brand illustration

FORMULA **Frame A2 + Atomic Delta Instruction**

"Replace only X with Y. Keep: camera angle, lighting, background, geometry."

RUNNABLE PROMPT B1: OBJECT SWAP (INPAINTING)

```
Edit this photo. Replace only the ceramic mug with a tall glass of iced latte with visible coffee and milk layers and ice cubes. Keep everything else exactly the same: camera angle, counter, espresso machine, background, light direction and shadows.
```

RUNNABLE PROMPT B2: LIGHTING & WEATHER (RELIGHTING)

```
Edit this photo. Change only the time of day to a rainy evening: dark dusk outside window, raindrops on glass, warm interior pendant lamps switched on. Keep exact same composition, objects, mug and camera angle.
```

In ChatGPT / Gemini: conversational delta prompt on top of frame.
In Leonardo / Kling: brush mask over mug ➔ replacement prompt.

Courier Master Sheet: Multi-Angle Generation Formula

A naive prompt generates a new random human in every scene. The engineering approach: first generate 4 angles on neutral grey and lock physical visual anchors.



Character Reference Sheet: 4 angles in neutral pose with studio lighting

5 Fixed Visual Anchors: short curly hair • small silver hoop earring • bright yellow jacket with grey stripe • turquoise backpack • white sneakers

NAIVE **"food courier cycling in the rain"**

The model generates a random face each time. Next scene: new age, facial structure, jacket. Consistency = 0%.

ENGINEERING **Character Sheet Generation (4 Angles on Neutral Grey)**

Studio lighting, neutral grey background, full body (front, 3/4, side profile, back). Master identity asset for all scenes.

RUNNABLE PROMPT FOR MASTER SHEET GENERATION (C1)

```
Character reference sheet on a plain light grey background: the same young woman
food delivery courier shown four times in a row - front view, three-quarter view,
side profile, back view - full body, standing neutral pose.
Fixed appearance: about 28 years old, short curly dark-brown hair, light-brown
skin, small silver hoop earring in left ear, bright yellow waterproof jacket with
reflective grey stripe across chest, black cargo trousers, white sneakers, square
turquoise thermal backpack. Studio photography, clean even lighting. No brand
names, no logos, no text.
```

Execution targets: Google Gemini, Midjourney, Leonardo, Kling. Save result as master `C1_sheet.png`.

Any Person as Reference: From 2–3 Photos to a Master Sheet

How to turn yourself or an actor into a controllable character: feed real photos to the model, enforce strict anti-beautification rules, and lock 4 canonical angles.

INPUT 1 • REFERENCES

2–3 real face photos



Frontal studio



Emotion / light



Daylight

INPUT 2 • CHAT / API PROMPT (GEMINI / CHATGPT / LEONARDO) 100% IDENTITY LOCK

Create a photorealistic character reference sheet based on attached real photographs of Ihar Rubanovich. The character is the exact same adult man, not a generic model.
Preserve authentic facial identity: exact face shape, forehead, eyes, eyebrows, nose, lips, jawline, hairline, skin tone, proportions. **Do not beautify, rejuvenate, stylize, or replace his face.**
Four horizontal angles: front view, three-quarter view, strict side profile, back view. Full body, standing neutral pose.
Outfit: bright yellow waterproof courier jacket with grey stripe, black cargo trousers, white sneakers, turquoise thermal backpack.
Studio lighting, light grey background. **No text, no logos.**

Execution: Drop these photos + prompt into Gemini / ChatGPT. In Python SDK: pass files to `contents`.



Generated Master Output: Facial identity 100% preserved across all 4 angles. Courier attire fitted without anatomical warping.

ANTI-PATTERN

Without `do not beautify/stylize`, models erase skin texture, alter nose shapes, and output generic glossy faces.

PRODUCTION BRIDGE

The resulting sheet is attached as a **Character Reference** across any storyline scene (rain, office, café) and into Kling for video.

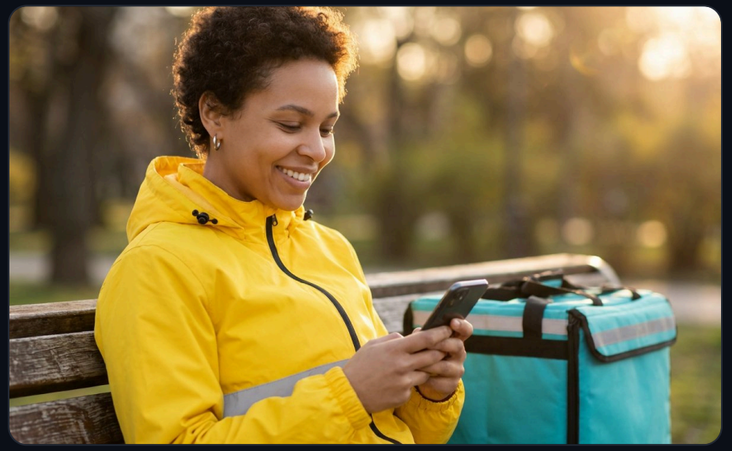
Courier Across Scenes & First Frame for Video (Kling / Veo)



Scene 1 (16:9): cycling in rain ➔ **base for video**



Scene 2: doorstep order delivery (35mm)



Scene 3: park bench break (85mm)

RUNNABLE SCENE TRANSFER PROMPT

```
Use attached character sheet (C1) as identity reference:
The same courier (same face, curly hair, silver hoop earring, yellow jacket with
grey stripe, turquoise backpack) is riding a city bicycle through a rainy
evening street with wet cobblestones reflecting shop lights. Medium shot, 50mm
lens, motion blur in background. Realistic photography. No logos, no text.
```

Aspect Ratios: 16:9 for landscape / YouTube, 9:16 for Reels / TikTok / Shorts (set via interface selector).

BRIDGE TO VIDEO **Pipeline to Kling 3.0 / Google Veo 3.1**

1. **Export frame:** take `C2_rain.png` (16:9) as keyframe (t_0).
2. **Image-to-Video:** upload into Kling or Veo **First Frame** slot.
3. **Motion prompt:** `"courier starts pedaling forward, wheels splash rain puddles, camera tracks along"`.
4. **Result:** 100% facial & outfit stability! Character does not morph in motion.

Tool controls: Gemini (multimodal attach) • Leonardo (Character Ref + Seed) • Kling/Veo (I2V First Frame).

One Packshot — Four Commercial Assets



Master Packshot: product on pure white

1. MASTER PACKSHOT PROMPT (D0)

```
Studio e-commerce packshot of a single handmade ceramic
mug: cylindrical body, speckled cream glaze with small
brown dots, unglazed terracotta-brown band at base, round
C-shaped handle on right. Centered, 3/4 view, pure white
background, soft even light. No text, no logo.
```



Lifestyle (D1): kitchen morning



Flat Lay (D2): top-down arrangement



In-Hand (D3): human scale perspective



Ad Banner (D4): promo hero poster

2. CATALOG SCENE PROMPT (WITH D0 REFERENCE)

```
Product photo session. Place exactly this mug (same shape, speckled cream glaze, terracotta base
band, handle on right) on light travertine kitchen counter next to morning window. Soft light,
50mm f/2.8. Do not change mug shape, clay glaze, or proportions.
```

Engineering control: Attaching reference `D0` + anti-drift negative constraints preserves shape and glaze 100%.

Explicitly Assign the Role of Each Image



Image 1 — Character (courier)



Image 2 — Product (mug)



Result: courier drinking from that exact mug in a morning cafe

ROLE RULE **Never say "combine these" (blurs inputs). Explicitly declare roles:**
Image 1 = identity reference (subject), Image 2 = product reference (item).

RUNNABLE COMPOSITION PROMPT (GEMINI NANO BANANA / CHATGPT)

```
Image 1 is the courier character, Image 2 is the exact ceramic mug.
The courier from Image 1, in her yellow jacket with grey stripe, sits at a coffee shop window
counter on her break and drinks from exactly the mug from Image 2 (same speckled cream glaze and
terracotta base). Turquoise backpack on floor next to stool. Morning light, 35mm lens, realistic
photography. No logos, no text.
```

Batch Generation via Code — Identical SDK to L03

```python
from google import genai
from google.genai import types
from PIL import Image

client = genai.Client()
ref_img = Image.open("packshot.png")

resp = client.models.generate_content(
    model="gemini-3.1-flash-image",
    contents=[ref_img, PROMPT_SPEC],
    config=types.GenerateContentConfig(
        response_modalities=["IMAGE"],
        image_config=types.ImageConfig(aspect_ratio="16:9")
    )
)
resp.parts[0].as_image().save("output_scene.png")
```

50 Product Cards in 3 Minutes

Replaces an entire day of manual browser clicking. Read a CSV database and iterate programmatic requests automatically.

> **10/90 Rule in Code:** The API call is 5 lines. The other 90% is catalog parsing, prompt templating, and automated artifact validation.

Technical Limitations & Production Boundaries

- **Anatomy & Tiny Text:** Fingers in crowds and dense paragraphs still suffer degradation.
- **Semantic Drift:** Successive multi-turn edits cause subtle drift in facial structure and logos.
- **Copyright Law:** Pure prompt generations without human authorship lack US/EU copyright protection.
- **Provenance Watermarks:** SynthID (invisible spectral watermark) and C2PA cryptographic metadata.



GREEN • AUTONOMOUS
Moodboards, concept art, internal presentation slides, storyboards.

YELLOW • HUMAN-IN-THE-LOOP
E-commerce cards, marketing banners, YouTube thumbnails. Human sign-off required.

RED • PROHIBITED
Legal contracts, medical packaging ingredients, third-party trademarks, non-consensual likeness.

PRACTICE • HOMEWORK ASSIGNMENT #4

Product Photoshoot or Delivery Courier Series

Track A: Commercial Product Photoshoot

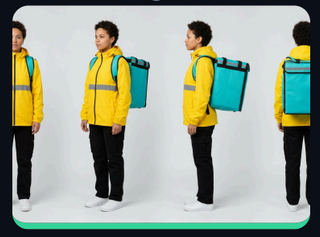
Pick a physical product on white background ➔ 4 commercial angles (lifestyle, flat lay, in hands, banner).

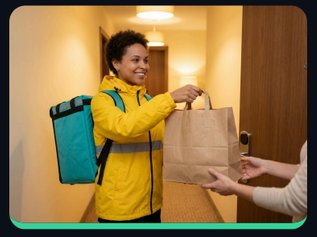





Track B: Glovo Food Delivery Courier

Character sheet ➔ 3 contextual scenes with visual anchors ➔ 1 targeted inpainting edit.






SUBMISSION REPOSITORY STRUCTURE

```
genai-homeworks/
└── L04/
    ├── images/ (01_free.png, 02_structured.png...)
    ├── prompts.md (prompts, tool, timestamp)
    └── qa.md (drift observations, what succeeded)
```

Required: Run one prompt across two tools (e.g. Leonardo vs. ChatGPT) and analyze the differences in `qa.md`. Invite collaborator: `ihar_rubanovich@epam.com`.

TAKEAWAYS • BRIDGE TO VIDEO

Four Principles of Visual AI Engineering

1. Tool to Match Task

Chat for concepts and text, SaaS for canvas and masks, API for batch automation.

2. Structure Over Freeform

6 explicit slots (Subject, Scene, Camera, Light, Style, Negatives) + Must-Have block.

3. Surgical Refinement Over Rerolls

Atomic "modify only X, preserve Y" commands outperform blind dice rolling.

4. Consistency via Visual Anchors

Master character sheet + 4–6 distinctive textural anchors preserve identity anywhere.



Lesson 05: Video Generation

Today's courier frames become the keyframes of tomorrow's video models (Kling, Veo, Runway). Our courier will start moving!